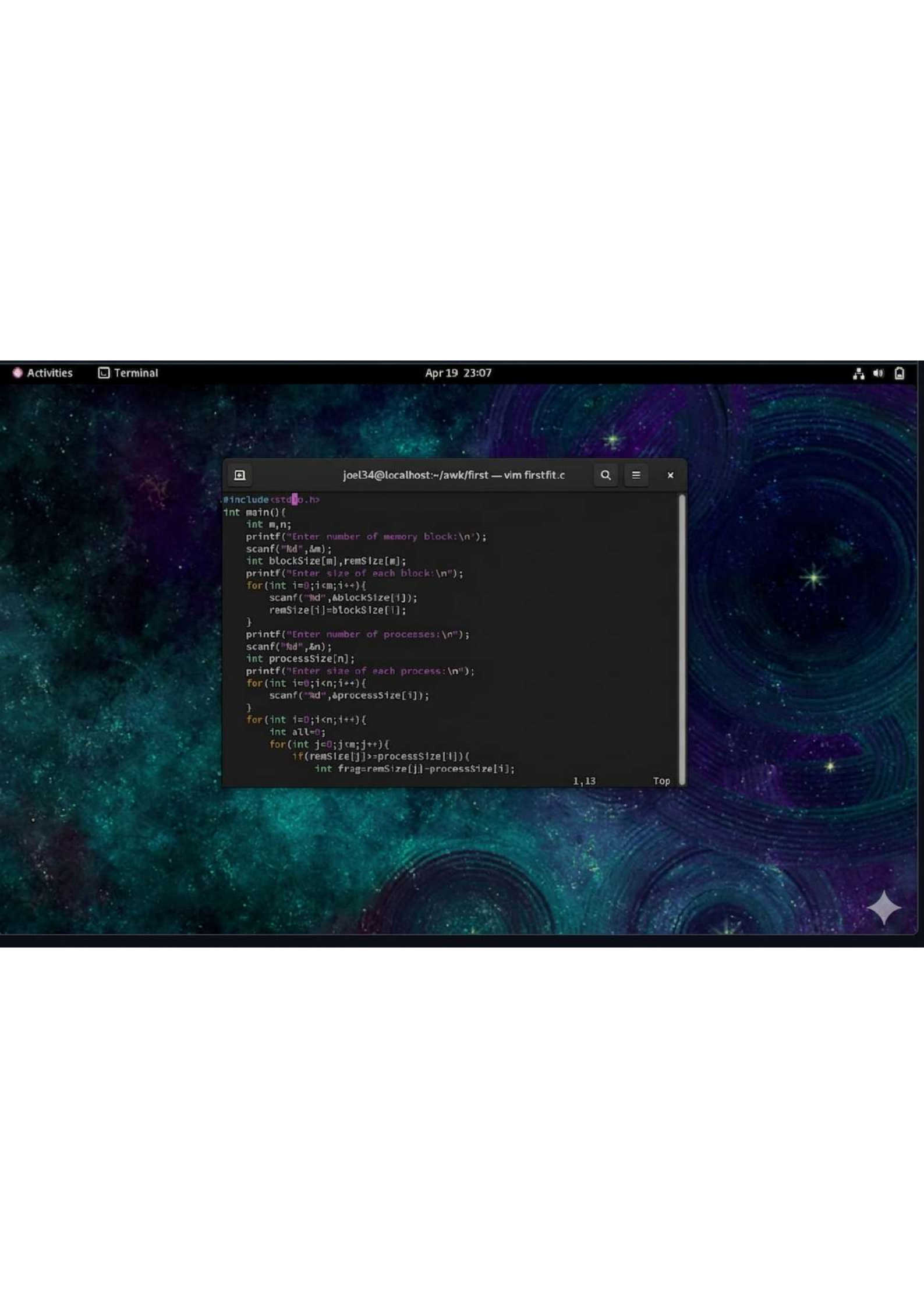




```
joel34@localhost:~/awk/first — vim firstfit.c
#include <stdio.h>
int main(){
    int m,n;
    printf("Enter number of memory block:\n");
    scanf("%d",&m);
    int blockSize[m],remSize[m];
    printf("Enter size of each block:\n");
    for(int i=0;i<m;i++){
        scanf("%d",&blockSize[i]);
        remSize[i]=blockSize[i];
    }
    printf("Enter number of processes:\n");
    scanf("%d",&n);
    int processSize[n];
    printf("Enter size of each process:\n");
    for(int i=0;i<n;i++){
        scanf("%d",&processSize[i]);
    }
    for(int i=0;i<n;i++){
        int all=0;
        for(int j=0;j<m;j++){
            if(remSize[j]>=processSize[i]){
                int frag=remSize[j]-processSize[i];
```

1,13 Top

```
joel34@localhost:~/producerconsumer — vim prod-cons.c
```

```
void *consumer(void *arg){
    int item;
    for(int i=0;i<BUFFER_SIZE;i++){
        sem_wait(&full);
        sem_wait(&mutex);
        item=buffer[out];
        printf("Consumed: %d\n",item);
        out=(out+1)%BUFFER_SIZE;
        sem_post(&mutex);
        sem_post(&empty);
    }
}

int main(){
    pthread_t p,c;
    sem_init(&empty,0,BUFFER_SIZE);
    sem_init(&full,0,0);
    sem_init(&mutex,0,1);
    pthread_create(&p,NULL,producer,NULL);
    pthread_create(&c,NULL,consumer,NULL);
    pthread_join(p,NULL);
    pthread_join(c,NULL);
    sem_destroy(&empty);
    sem_destroy(&full);
}

-- INSERT --
```

41,2-5 82%

```

    item=buffer[out];
    printf("Consumed: %d\n",item);
    out=(out+1)%BUFFER_SIZE;
    sem_post(&mutex);
    sem_post(&empty);
}

int main(){
    pthread_t p,c;
    sem_init(&empty,0,BUFFER_SIZE);
    sem_init(&full,0,0);
    sem_init(&mutex,0,1);
    pthread_create(&p,NULL,producer,NULL);
    pthread_create(&c,NULL,consumer,NULL);
    pthread_join(p,NULL);
    pthread_join(c,NULL);
    sem_destroy(&empty);
    sem_destroy(&full);
    sem_destroy(&mutex);

    return 0;
}

```

-- INSERT --

31,1 Bot

```
joel34@localhost: ~/producerconsumer
prod-cons.c:(.text+0x1b6): undefined reference to `pthread_create'
/usr/bin/ld: prod-cons.c:(.text+0x1f8): undefined reference to `pthread_join'
collect2: error: ld returned 1 exit status
[joel34@localhost producerconsumer]$ vim prod-cons.c
[joel34@localhost producerconsumer]$ gcc prod-cons.c -o prod-cons
[joel34@localhost producerconsumer]$ ./a.out
bash: ./a.out: No such file or directory
[joel34@localhost producerconsumer]$ ./prod-cons.c
bash: ./prod-cons.c: Permission denied
[joel34@localhost producerconsumer]$ ./a.out prod-cons.c
bash: ./a.out: No such file or directory
[joel34@localhost producerconsumer]$ gcc prod-cons.c
[joel34@localhost producerconsumer]$ ./a.out prod-cons.c
Produced: 0
Produced: 1
Produced: 2
Produced: 3
Produced: 4
Consumed: 0
Consumed: 1
Consumed: 2
Consumed: 3
Consumed: 4
[joel34@localhost producerconsumer]$
```